

Four minutes, **ten slides.**

What to say on every slide, **29 September.**

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Run of show · 29 Sept

Four minutes, ten slides, about 770 spoken words, 5.1 minutes at 150 words a minute. The spine: **self, us, now** (02, 03), then **why, how, what** (04, 05, 06), then **who else, where it goes, the ask** (07–10). 07 and 08 are the fastest stretch: one beat each.

SLIDE	CLOCK	SEC	CLICKS	BEAT	ONE LINE TO LAND
01 Control is an illusion	0:00–0:12	12	1	hook	Confidence without control.
02 Founder	0:12–0:40	28	1	self	This is the thing I kept needing and nobody sold.
03 Do my cells look normal?	0:40–1:05	25	1	us, now	Cells are the input nobody certifies.
04 One lab, or six?	1:05–1:45	40	3	why now	The model isn't the hard part. The reference is.
05 It passed every check	1:45–2:20	35	3	how	Your own past is not a reference. The field is.
06 The product	2:20–2:45	25	1	what	Four months becomes one day.
07 Identity, yes. Behaviour, no.	2:45–2:55	10	1	who else	The other half of the certificate.

SLIDE	CLOCK	SEC	CLICKS	BEAT	ONE LINE TO LAND
08 Three buyers, one certificate	2:55–3:10	15	1	who buys	The labs that build the reference use it free.
09 The ugliest dataset in biology	3:10–3:35	25	1	the build	Name the failure, not just flag it.
10 Every plate, placed in the field	3:35–4:00	25	0	vision, ask	Have images of U2OS cells? Talk to me.

Captions stay hidden while you present (Tweaks › captions); the shared link and PDF show them. If you're late at 2:20, cut the "why now" sentence on 04 and the "who pays" line on 06.

01 Control is an illusion (0:00–0:12)

On screen while you're introduced. Two lines drop from the top like dive lines: green is the field's datum, pink is yours, and the gap between them is the product. Let the dial drift.

0:00 Hi, I'm Kat. Every result in biology is a difference from a control, and nobody checks the control. So I'm building confidence without it. *Point at the pink dot, then the green ring. Click.*

02 Founder (0:12–0:40)

Self. Let the logos and the pink line under your name do the credential work.

0:12 Five labs, MGH, AbbVie, MIT, insitro, Axiom, and the same question at every bench: do my cells look normal? Nobody could answer it, and it made the science slow and wrong.

0:20 At Axiom we onboarded primary hepatocytes through a good CRO. Their checks passed; ours passed. We had noticed some uneven spreading in wells, and

nothing could tell us whether it mattered. The first sign it did came from outside: the Broad couldn't align our profiles to their reference. That's the gap. Not seeing something; knowing whether it matters.

0:35 This is the thing I kept needing and nobody sold. *Click.*

Founder detail, for the judge question (about 30 seconds). Three methods, three papers, one thread. At AbbVie I built a neuronal tau aggregation assay for drug discovery [3]: the readout was what the cells did, not a label on them. At Axiom I built Cell Painting in primary human hepatocytes for cytotoxicity and mode of action [1], ran assay development for two of the largest Cell Painting datasets in the world, and invented two bioactivity detection methods that see a small molecule's effect on cells at about ten times lower concentration than LDH, ATP or cell count. With the Carpenter lab we showed that cell count alone predicts small-molecule bioactivity benchmarks [2]: even the simplest phenotype carries the signal. The thread is that bioactivity is a measurement of behaviour, and behaviour is the half of QC nobody measures. Every lab already images its control wells; Datum Line turns that image into the certificate, against the field.

Precision to hold. The question predates plates: at MGH it was whether a *Xenopus* oocyte, a single frog egg cell, looked normal before a recording. Say only what your author contribution statements support: "built" on [1] and [3], "we showed" on [2]. The two bioactivity methods and the ten-fold sensitivity are from your resume (macrofluidic.me/resume); if asked, name the comparison assays and the concentration range. The 2019 GABA_A paper [5] is the MGH anesthesia lab, Harvard Medical School, 2017 to 2018. The industry claim is the deck's own: STR and vendor certificates cover identity, bioactivity covers behaviour, and behaviour is what a control well measures in any lab that images one.

Precision to hold. "Published with" lists coauthor institutions, never partners. The Axiom story has no villain: the CRO passed every check it had, Axiom passed every check it had, and the failure was that no check for this existed. Name no vendor and no person. The uneven distribution was noticed; its severity and its effect on data

quality were unknown, so never present it as the cause, and never say "spun down unevenly". If asked what it was: "We still don't know for certain. The uneven spreading is one hypothesis; whole-well images show it more easily than our fields, and a field reference is what would tell you whether it matters." The sequence is: local checks passed, uneven spreading noticed but unquantified, the Broad flagged that reference profiles would not align, the cause is still open. Say what happened, not how it was fixed. Confirm with Axiom that you can tell it; if not, say "at one lab".

03 Do my cells look normal? (0:40–1:05)

Us, then now. Point at the pink slice, then the six search results, then the two edges of card 3: run it, or hold.

0:40 This isn't my problem; it's all of ours. About twenty-eight billion dollars a year of US preclinical research can't be reproduced, and the biggest slice, about a third, is inputs.

0:52 Today you search, you ask around, then you guess. I have seen both edges. One lab ran it anyway: hundreds of thousands in R&D before the SOP was fixed. Another held: months developing cells that were already good enough. One too fast, one too slow. Cells are the input nobody certifies. Think growth chart: nobody plots their cells against the population. *Click.*

Precision to hold. \$28B is a 2015 US estimate; the 36% covers all reagents and reference materials, not only cells. The search card shows six real U2OS images from six papers, 2009 to 2023, via IDR and JUMP; the sources are in the Fig. 1 caption. Card 3 names neither lab and gives no exact figure; keep it that way on stage. The \$2.6M screen estimate stays on 06 and in the appendix. The why-now block on the slide cites JUMP [4]; the spoken why-now stays on 04.

04 One lab, or six? (1:05–1:45, three clicks)

Why now. State 1 is a game: let the room look for four seconds.

1:05 Six untreated wells. Same line, same protocol, no drug. One lab, or six? *Beat. Click.*

1:15 Six labs. Nobody here can tell; a classifier can: 99.9 percent of wells, against nine by chance, on plates it never saw. The pink one is the lab we follow. *Click.*

1:28 Why now: this data is public, eleven labs, free, and embeddings are a commodity. Same-size nuclei, different texture: that's what moves the point. The model isn't the hard part. The reference is. *Click.*

Precision to hold. Eleven labs, 93,228 untreated wells, 1,872 plates, held out by plate; 78 wells wrong. Holding out whole batches gives 99.6%; four labs on the same microscope model, 99.99%. The signature is in every feature family, image-quality features included, so it reflects the whole imaging pipeline as well as the cells. Say "the lab", not "the cells". Cell Painting fluorescence; brightfield is the bet.

05 It passed every check (1:45–2:20, four views)

How. Dark slide: slow down. Four views of one lab.

1:45 One lab, thirteen batches, June to October. Cells per well stay in range. On 21 June they dip, and the lab can see that. After that, nothing to see. *Click.*

1:52 The same batches, placed against the field. At the end of August the pink line crosses the green: this lab is now further from its own June baseline than from another lab. By October, 8.1 spreads from itself and 5.1 from lab 8. Every well is still recognisably lab 2. *Click.*

2:02 Here is the map: every untreated well, one shade of green per lab. June and October are different places. Two plates of this lab sit seven times further apart after thirteen weeks than inside one batch. Your own past is not a reference. The field is. *Click.*

Fourth click, if there is time. The same drug, a different answer. Every point is one batch of one lab. The further a batch has drifted from its lab's baseline, the less its answers to the same drugs agree with that baseline: rho minus 0.71, 118 batches, eleven labs. Lab 2's agreement with its own June answer: 0.85 before the flag, 0.67 after. That is the failure I lived: profiles that would not align to the field's reference. *Click.*

For ML people. The line is on the board in the second state of 04. Centring each plate on its own controls takes lab identity from 99.9 to 24 percent: right for discovery, wrong for QC. On the fourth click of 05, the correlation is inside labs, after removing each lab's mean. In lab 2 the flagged batches are also the last three, so inside that lab the flag and the calendar cannot be separated. The crosses are batches whose positive controls show no effect (A8); they are not drift.

06 Know on day one, not month four (2:20–2:45)

What. Read the sentence under the headline. Then the strip, left to right: thirteen uploads. Then the certificate for the first flag.

2:20 The product: the control-well images you already take. They stay yours. We place them against the field. One answer in a day: in or out, and which features moved.

2:35 Here it is, replayed on lab 2. Thirteen batches, each judged with only what was known that day. The first sets the baseline. Nine come back in distribution. On 30 August it says outside: five weeks before the October batch was run. And it says where to look: ER and AGP stain intensity, up almost three standard deviations. Cell count unchanged. Four months becomes one day. *Click.*

Precision to hold. The replay is hindsight on public data, against the field as published. The flagged batches hold 5, 6 and 14 plates. The certificate says where to look, not why. Across the field 6 of 129 batches flag, and lab 2's verdicts hold under nine of ten variants (A11, A12). Brightfield is still the bet (A10).

07 Identity, yes. Behaviour, no. (2:45–2:55)

Point at the open tick in the Datum Line column: identity is done with a partner, STR. One sentence.

2:45 STR proves which line you have. Nothing proves it behaves like everyone else's, or says which way to move it back. STR is a collaborator, not a rival. *Click.*

08 Three buyers, one certificate (2:55–3:10)

Who buys. Walk the three columns; the pink row is what they pay.

2:55 Three buyers. Cores risk the paper; they contribute control images and use it free, and that builds the reference. Pharma and CROs risk the screen; they pay per cell line per quarter, about a hundred and seventy times less than one unchecked screen. Cell banks risk the lot; they pay per certified lot, and that builds the network. Proposed, not yet validated. *Click.*

Precision to hold. The prices are the README's proposal, not a validated model: free for contributors, ~\$5–15k per line per quarter for screening groups, a fee per certified lot. If you need ten seconds back, say only the first sentence of each column.

09 The ugliest dataset in biology (3:10–3:35)

The build. This is the blank-cheque answer, said before anyone asks. Point at the six real wells, then the four corrupted tiles.

3:10 If someone wrote the cheque, I'd build the ugliest dataset in biology, on purpose. These six wells are the same line, same protocol, no drug, and six labs already look like this: five microscope models, eight plate types. Then plant the failures: defocus, exposure, clipped wells, bad metadata.

3:23 Every method has to name which thing went wrong, not just that something did: a public benchmark, name the failure. A foundation other people can stand on, instead of Google Images. *Click.*

10 Every plate, placed in the field (3:35–4:00)

Vision, then the ask. Point at the grid: every batch of every lab. Then the right card.

3:35 That's the vision, and here it is on public data: 129 batches, eleven labs, every one replayed. Six sit outside their own baseline. In four, the known answer did not come out. Every certificate adds a plate to the reference, and the reference is the moat: nobody can build it alone.

3:47 What I need: images, U2OS first; ML people better than me; bench time; and a pre-seed to run it. If you run a cell line somebody else also runs, you already have the data.

3:57 Have images of U2OS cells? Talk to me. *Stop.*

Precision to hold. Price and test-set budget are not set: say "per line, per quarter" and "the proof of concept sets the size". Don't name a partner lab until one has agreed. Name no JUMP lab beyond its number. The four failed batches are one lab's, 65 plates; the data cannot say whether the compounds were left out or the plate map is wrong, so say neither.